

Joel Beauregard

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EDUCATION

University of Texas at Arlington

Arlington, TX

B.S. in Computer Science, Cybersecurity Certificate, GPA: 4.0

May 2027

EXPERIENCE

Software Engineering Intern

May 2026 – Aug. 2026

Oracle - OCI

- Architected & developed an **AI reporting agent** automating customer onboarding requests from ticket to PR
- Implemented **SparkSQL** validation with query generation from ticket-provided business logic when needed
- Designed a persisted **state-machine orchestrator** enforcing workflow execution, validation, and approval gates
- Automated **20+ live customer requests**, reducing report onboarding time by **~94%**
- Built & deployed **3 MCP tools** for database, table, and schema discovery across hot and cold storage
- Enabled data discovery, investigations, detector development, and schema-grounded **SparkSQL** generation
- Built resilient catalog caching with scheduled refresh and last-good snapshot fallback

Software Engineering Intern

May 2025 – August 2025

Oracle - OCI

- Architected & documented a new Network Traffic Analytics feature for VCNs in Oracle's **PCA** Product
- Evaluated network traffic flows and system constraints to guide architectural decisions for feature design
- Used **Python, Bash, Kubernetes, Cisco Switches, and Podman** to develop and validate feature components
- Verified feature integration within existing network infrastructure to ensure compatibility with VCN architecture
- Integrated into **CI/CD workflows** that provision **Kubernetes** containers and run unit tests for validation
- Containerized Network Controller unit tests for **180+ tests** utilizing a **MySQL** server container
- Improved unit test performance by **80%** and enhanced error handling by refactoring **50%** of tests

Research Assistant

August 2024 – December 2024

Tacit Captions – UTA / The Hybrid Atelier

- Collaborated with a team of six to develop a multi-modal captioning system and co-author a research paper
- Developed **Python (pandas)** pipelines for time-series sensor data to generate synchronized WebVTT captions
- Reviewed existing multi-modal and assistive captioning systems to inform system design
- Authored technical sections detailing signal processing and sensor data extraction workflows

PROJECTS

DevShell | *Linux, Bash, Python, Docker, Cloud Infrastructure*

2026 – Present

- Developing a managed **Linux** development platform to eliminate local environment setup overhead
- Architecting a university-scale infrastructure model with horizontally scalable compute nodes (~35 students/node)
- Implementing cloud networking architecture using **VPCs** and **Network Load Balancers (NLBs)**
- Provisioning isolated per-user environments with **SSH** access, resource quotas, and automated environment setup
- Engineering persistent storage systems (**NFS**-backed) with enforced per-user disk quotas (8–10GB)
- Extending the platform with a professor-facing interface for assignment distribution and **CI/CD** via **Gitea**

Concurrent TCP Port Scanner | *Python, Asyncio, Scapy*

- Designed and implemented a **concurrent** TCP port scanner to scan multiple hosts and large port ranges
- Implemented SYN-based scanning to identify open ports while minimizing connection overhead
- Built banner-grabbing functionality to extract service metadata from open ports
- Developed an **asynchronous** scanning pipeline to improve throughput
- Structured modular components to support extensibility

TECHNICAL SKILLS

Programming: Python, Bash, Java, MySQL, C, C++

Technologies: Podman, Docker, OCI, Kubernetes, Helm Charts, Terraform

Tools/Systems: Ubuntu, Git, VCNs, Cisco NX-OS, SQLAlchemy, Scapy, Asyncio, pandas

Interests: Networking, Distributed Systems, Cloud Computing, Cybersecurity, Artificial Intelligence